

CAROLINA LOBATO

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Düsseldorf, North Rhine-Westphalia, Germany

Education

PhD Molecular Biosciences and Biotechnology

2021 – 2025

Graz University of Technology 

Austria

Thesis: [Microbiome-Driven Sustainability in Cannabis Seeds](#), defended with distinction

MSc Molecular Biomedicine

2014 – 2016

University of Aveiro 

Portugal

Thesis: [Alternative Initiation Codons in Candida cylindracea](#)

BSc Biotechnology

2009 – 2014

Polytechnic University of Coimbra 

Portugal

Internship LLLP-Erasmus

University of Veterinary Sciences Brno (Czechia) 

Professional Experience

Postdoctoral Researcher

2025 – present

Heinrich-Heine University Düsseldorf 

Germany

with Prof. Dr. Björn Usadel

Graduate University Assistant

2021 – 2024

Graz University of Technology 

Austria

with Univ.-Prof. Dipl.-Biol. Dr.rer.nat. Gabriele Berg

Research Assistant

2020

Graz University of Technology 

Austria

with Univ.-Prof. Dipl.-Biol. Dr.rer.nat. Gabriele Berg & Dr. Tomislav Cernava

FCT Research Fellow

2017 – 2019

University of Lisbon

Portugal

with Dr. Leonor Morais-Cecílio  & with Dr. José Carlos Rodrigues & Dr. Ana Alves 

Undergraduate Research Intern

2013 – 2014

Polytechnic University of Coimbra 

Portugal

with Dr. Sandra Gamboa

Funding & Awards

• Early Career Grant

2026

HHU Düsseldorf Strategic Research Fund (SFF)

• Seal of Excellence, score: 96.6/100

2026

Horizon Europe MSCA Postdoctoral Fellowships

Mentorship

Supervision of 4 Master's and 4 Bachelor's students, including experimental design, molecular and microbiological techniques, and data analysis.

Teaching

• Lecturer at HHU Düsseldorf

2026

– Applied Microbial Ecology & Sustainability (M-2216)

• Teaching Assistant at HHU Düsseldorf

2026

– Seed Biology (V-525)

• Guest Lecturer at TU Graz

2023–2024

– Environmental Microbiology (MOL.963UF)

• Graduate Teaching Assistant at TU Graz

2021–2024

– Microbiome Analysis (MOL.972UF)

- Environmental Biotechnology (MOL.935UF)
- Biodiversity and Applied Microbiology (MOL.731UF & PLA.221UF)
- Biotechnology (CHE.177UF)
- Environmental Microbiology (MOL.963UF)
- Microbiology (MOL.209UF)
- Bioinformatics (MOL.923UF)

- **Undergraduate Preceptor** at IP Coimbra
 - Genetic Engineering

2013

Professional Development & Service

Co-founder & Coordinator

- Seed Microbiome Working Group (SMWG) 

2022–present

Conference Organization

- Session Organizer at SeedCon26 
- Session Chair at SeedCon26 
- Session Chair at Theodor Escherich Symposium 

2026

2022–2024

Memberships

- International Society for Microbial Ecology (ISME) 
- Austrian Association of Molecular Life Sciences and Biotechnology (ÖGMBT) 

Peer Review

Environmental Microbiome, Scientific Reports, Frontiers in Microbiology, Plant and Soil

Training

- Genome Reconstruction from Metagenomic Data (CEPLAS/UoC) 2025
- Reproducible Research with R and Quarto (ETH Zürich) 2024
- University Didactics Certificate (TU Graz) 2022
- Metagenomics Training School (ARI/CyI/CA16110) 2021
- Command Line Certificate (Code Academy) 2021
- Metagenomics Applied to Surveillance of Pathogens and AMR (DTU/Coursera) 2020
- Programming Initiation for Biologists Certificate (EPCV/Portuguese Order of Biologists) 2019
- Cytogenetics and Genomics in Diagnosis and Research (CIMAGO/FMUC) 2018

Community Outreach

- **Speaker** at TU Graz [Science Day](#) 2023
- **Organizer**
 - TU Graz [Long Night of Research](#) 2024
 - Ciência Viva outreach program [PEC 282](#) 2014
- [Research blog](#) and [science communication](#) ongoing

Research Output

Peer-reviewed Publications

- Chen X., Olimi E., **Lobato C.**, Cernava T. & Berg G. *mSystems*, 2025
[The tomato seed microbiome is mainly shaped by host genotype and production site.](#)
 Defined a seed core microbiome in tomato and predicted host and environment links to microbial composition.
- Olimi E., Wuggenig R., **Lobato C.**, *et al.* *Environmental Microbiology Reports*, 2025
[Insights into the *Mondia whitei* microbiome across geographic regions in Eastern Africa.](#)
 Revealed location-specific microbial and metabolomic signatures with implications for bio-prospecting and conservation of natural resources.
- Lobato C.**, Abdelfattah A., Berg G. & Cernava T. *Environmental Microbiome*, 2025
[Defining the cultured and uncultured bacterial fractions in *Cannabis* seeds.](#)
 Demonstrated the role of microbial abundance and interactions in culturability, highlighting strategies to recover uncultured taxa.
- Lobato C.**, Freitas J.M., Habich D. *et al.* *Microbiome*, 2024
[Wild again: Recovery of a beneficial *Cannabis* seed endophyte from low domestication genotypes.](#)
 Showed that low-domestication genotypes harbor higher diversity and retain beneficial microbial associations lost during domestication.
- Abdelfattah A., Tack A., **Lobato C.**, *et al.* *Trends in Microbiology*, 2023
[From seed to seed: The role of microbial inheritance in the assembly of the plant microbiome.](#)
 Proposed a conceptual framework describing seed microbial transmission across plant life stages and its evolutionary implications.

6. Inácio V., **Lobato C.**, Graça J. & Morais-Cecílio L. *Tree Physiology*, 2021
[Cork cells in cork oak periderms undergo programmed cell death and proanthocyanidin deposition.](#)
Identified key molecular and cellular processes underlying cork formation and stress responses.

Selected Presentations

- **Invited Speaker** *Genetic control of the seed microbiome: Implications for sustainable cropping.* SeedCon Potsdam (2026)
- *Symbiotic Futures: Leveraging Seed Microbiomes for Sustainable Cannabis Cultivation.* LANDSCAPE Berlin (2024)
- *Healthy Cannabis Seeds Harness Geno- and Chemotype-Specific Microbial Signatures.* SMNS Maribor (2023)
- *Exploring the Cannabis Seed Microbiome Through Domestication.* Young AMICI Symposium (2023)
- *Characterization of the Cannabis Seed Microbiome.* Virtual Cannabis Research Conference (2021)

Selected Posters

- ICC, Málaga (2026)
- Plant Microbiome Symposium, Amsterdam (2024)
- BAGECO, Copenhagen (2023)
- ISS Congress, Lyon (2022)
- miCROPe Symposium, Vienna (2022)

Additional Information

Languages: Portuguese (native), English (proficient), Spanish (intermediate), German (A2)

Programming: R, Bash, Python (basic)

Driving License: B